

Lab 3 - Report

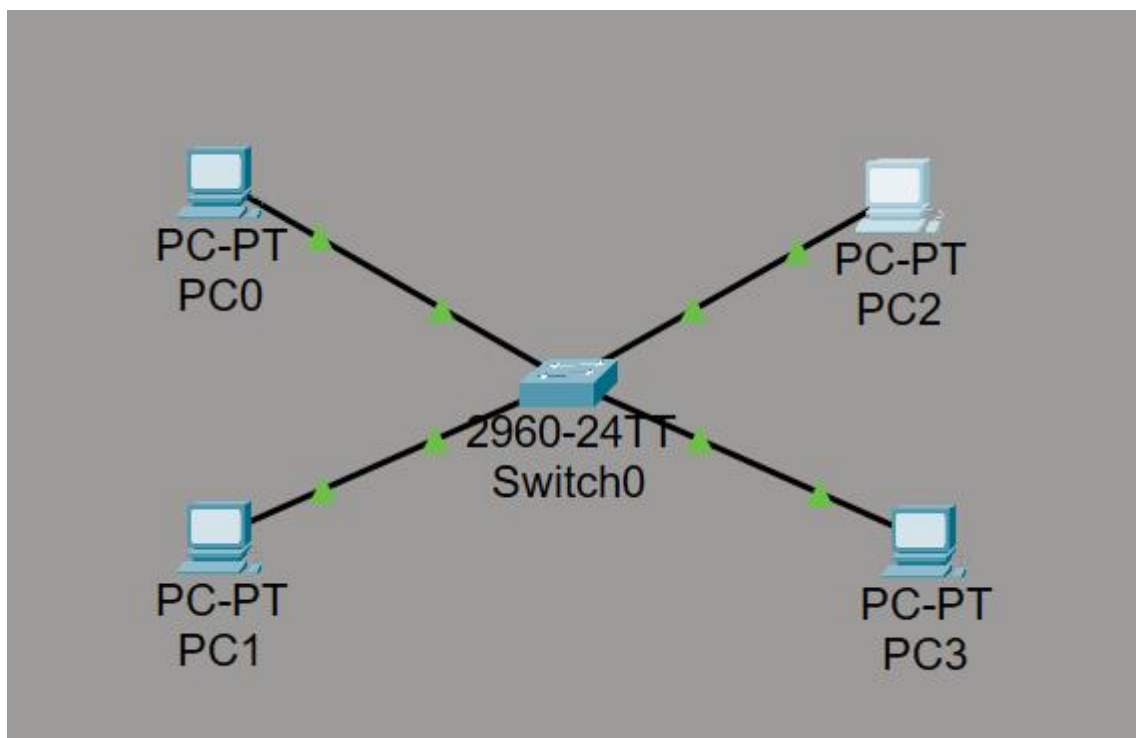
Student name: Bakhramov Abdulaziz

Student ID: 24B031692

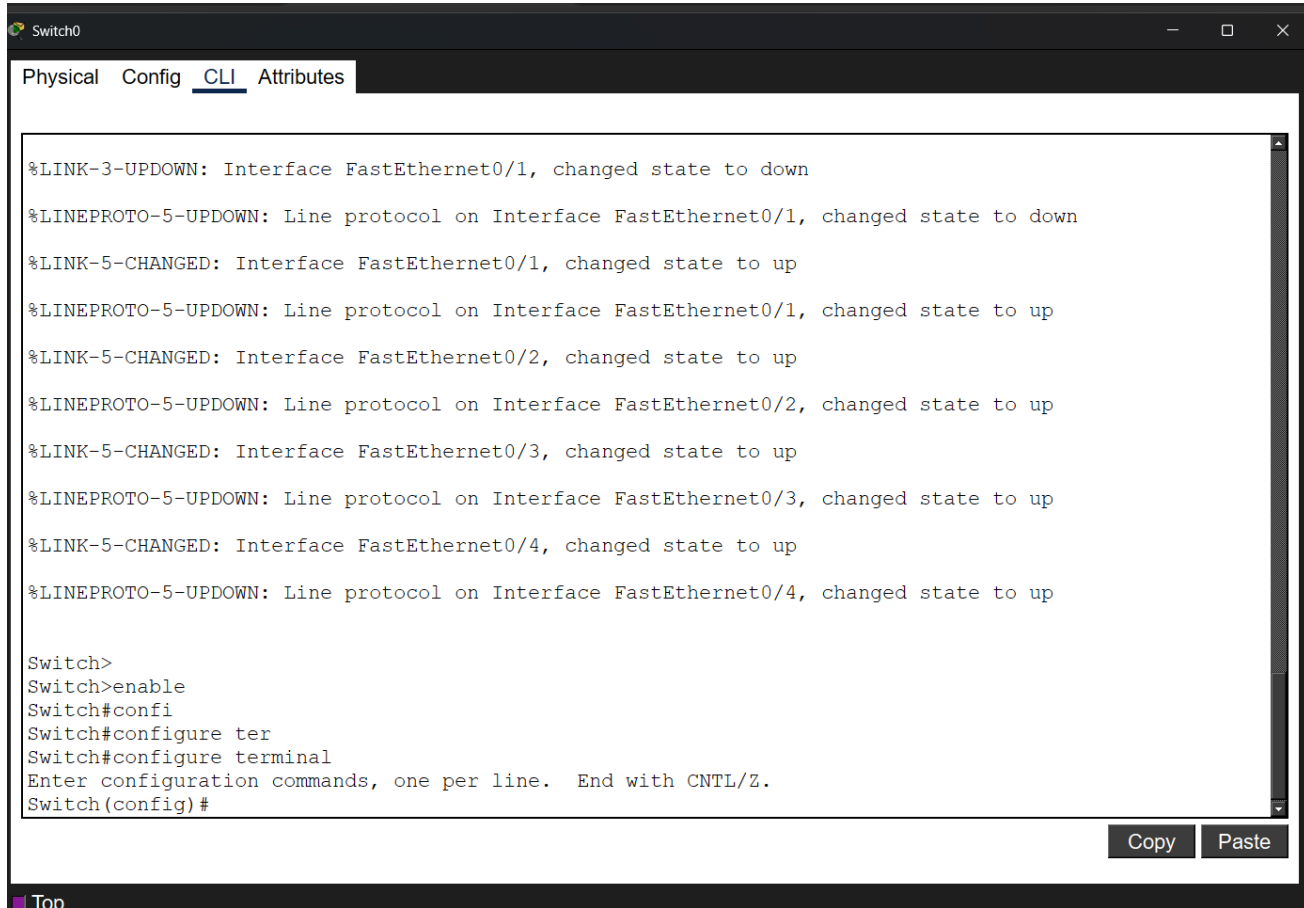
Group: Mon 11:00-13:00

VLAN Technology

1.Part 1. Access port



2. Access the CLI of the switch and enter the configuration mode:



The screenshot shows a network switch CLI window titled "Switch0". The interface has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" currently selected. The main display area shows a series of status messages for various interfaces (FastEthernet0/1, FastEthernet0/2, FastEthernet0/3, FastEthernet0/4) indicating link and line protocol state changes (UPDOWN, CHANGED). Below these messages, the CLI prompt "Switch>" is shown, followed by the commands "enable", "configure", and "terminal". The prompt then changes to "Switch(config)#". At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. A "Top" button is located at the bottom left of the window.

```
Switch0
Physical Config CLI Attributes

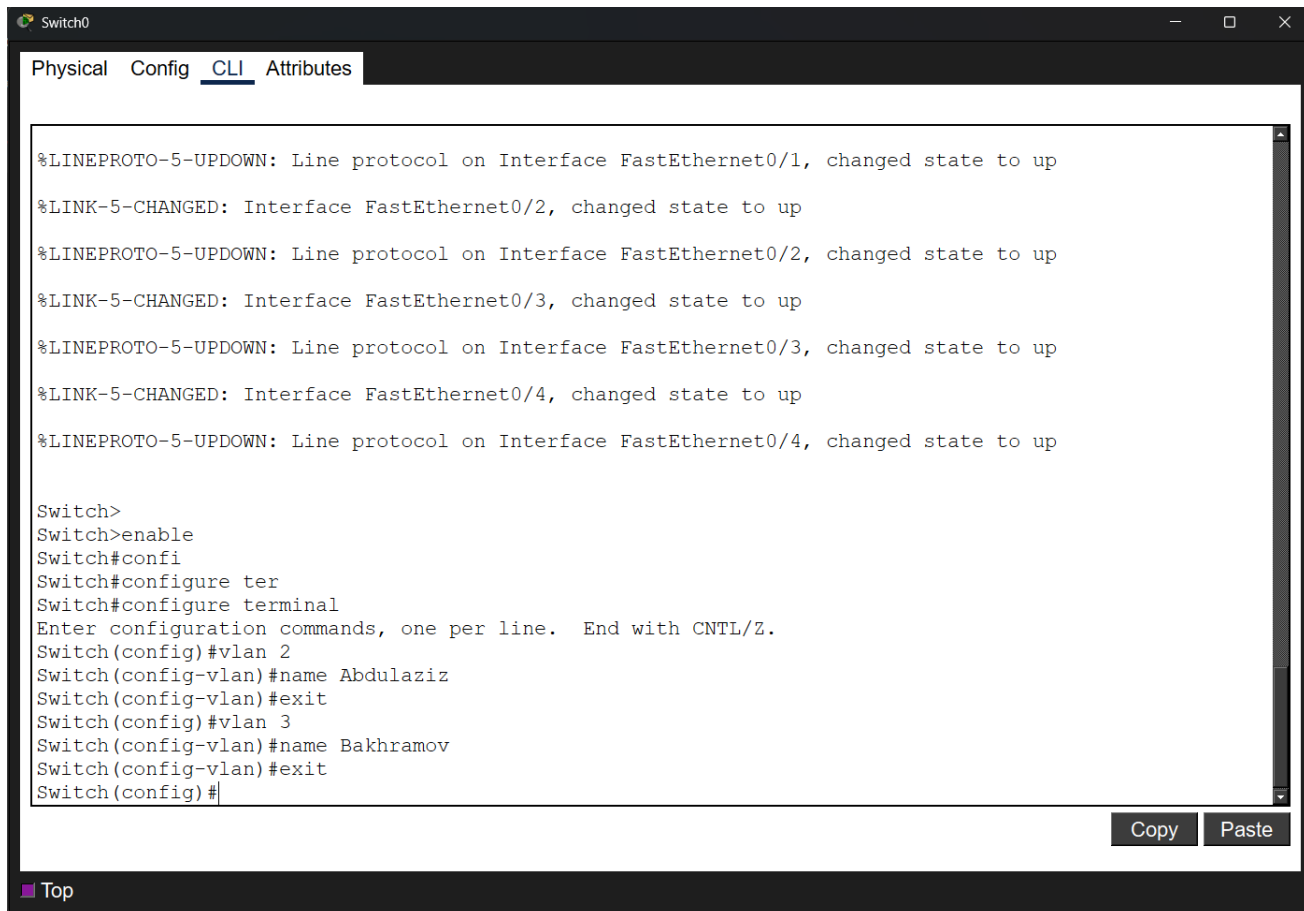
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch>
Switch>enable
Switch#confi
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
```

Copy Paste

Top

3. Create Vlans



The screenshot shows the same network switch CLI window as before, but now displaying the configuration of two VLANs. The status messages for interfaces are still visible at the top. Below them, the CLI prompt "Switch>" is shown, followed by the commands "enable", "configure", and "terminal". The prompt then changes to "Switch(config)#". The user enters "vlan 2", followed by "name Abdulaziz", and "exit". The prompt changes to "Switch(config-vlan)#". The user then enters "vlan 3", followed by "name Bakhramov", and "exit". The prompt changes to "Switch(config-vlan)#". At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. A "Top" button is located at the bottom left of the window.

```
Switch0
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch>
Switch>enable
Switch#confi
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name Abdulaziz
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name Bakhramov
Switch(config-vlan)#exit
Switch(config)#
```

Copy Paste

Top

4. Configuring interfaces

```
Switch(config-vlan)#name Abdulaziz
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name Bakhramov
Switch(config-vlan)#exit
Switch(config)#interface fa
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#sw
Switch(config-if)#switchport mode access
Switch(config-if)#swi
Switch(config-if)#switchport mode access vlan 2
Switch(config-if)#^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport access vlan 2
Switch(config-if)#interface fa
Switch(config-if)#interface fa
Switch(config-if)#interface fast
Switch(config-if)#exit
Switch(config)#interface fa
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#sw
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

Copy Paste

Physical Config CLI Attributes

```
%SYS-5-CONFIG_I: Configured from console by console

Switch#vlan 3
Switch#^
% Invalid input detected at '^' marker.

Switch#inter
Switch#conf
Switch#configure
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#in
Switch(config)#interface faste
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#sw
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#inter
Switch(config)#interface fas
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#swi
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

Copy Paste

5. Check VLAN configuration

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
2	Abdulaziz	active	Fa0/2, Fa0/3
3	Bakhramov	active	Fa0/1, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch#

6. Save the configuration to memory

```
Physical Config CLI Attributes
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan br
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
2    Abdulaziz              active    Fa0/2, Fa0/3
3    Bakhramov               active    Fa0/1, Fa0/4
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
Switch#wr men
^
% Invalid input detected at '^' marker.

Switch#wr mem
Building configuration...
[OK]
Switch#
```

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7. Configure IP addresses for all PCs.

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address 192.168.2.1

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:16FF:FE60:373E

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Top

8. Verify with ping

```
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

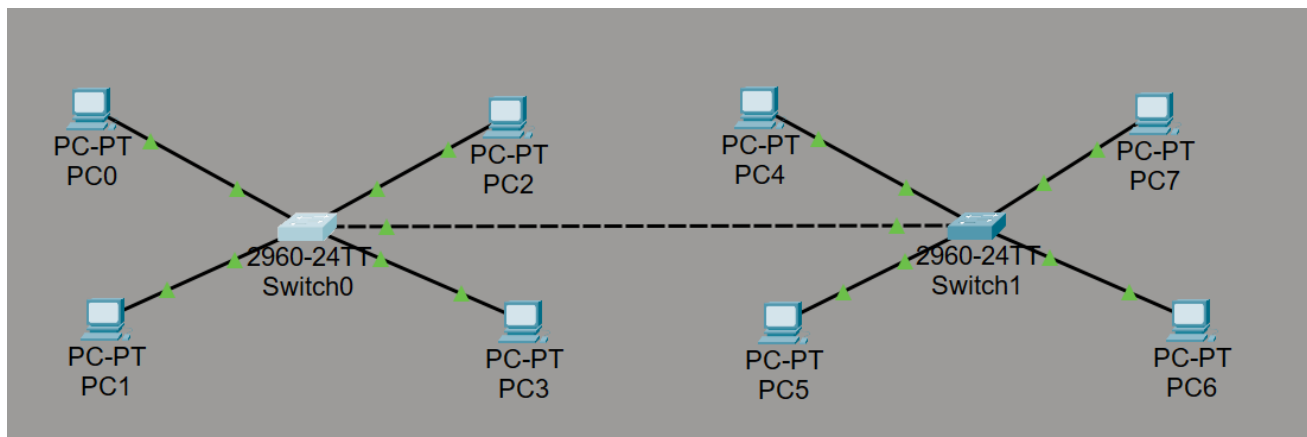
Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=8ms TTL=128
Reply from 192.168.2.1: bytes=32 time=4ms TTL=128
Reply from 192.168.2.1: bytes=32 time=4ms TTL=128
Reply from 192.168.2.1: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 5ms

C:\>
```

1.Part 2: Trunk port's topology



2. Configure Trunk Port on the GigabitEthernet0/1 Interface.

```
Switch>swit
Switch>configure ter
Switch>enable
Switch#confi
Switch#configure
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interfa
Switch(config)#interface giga
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode th
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

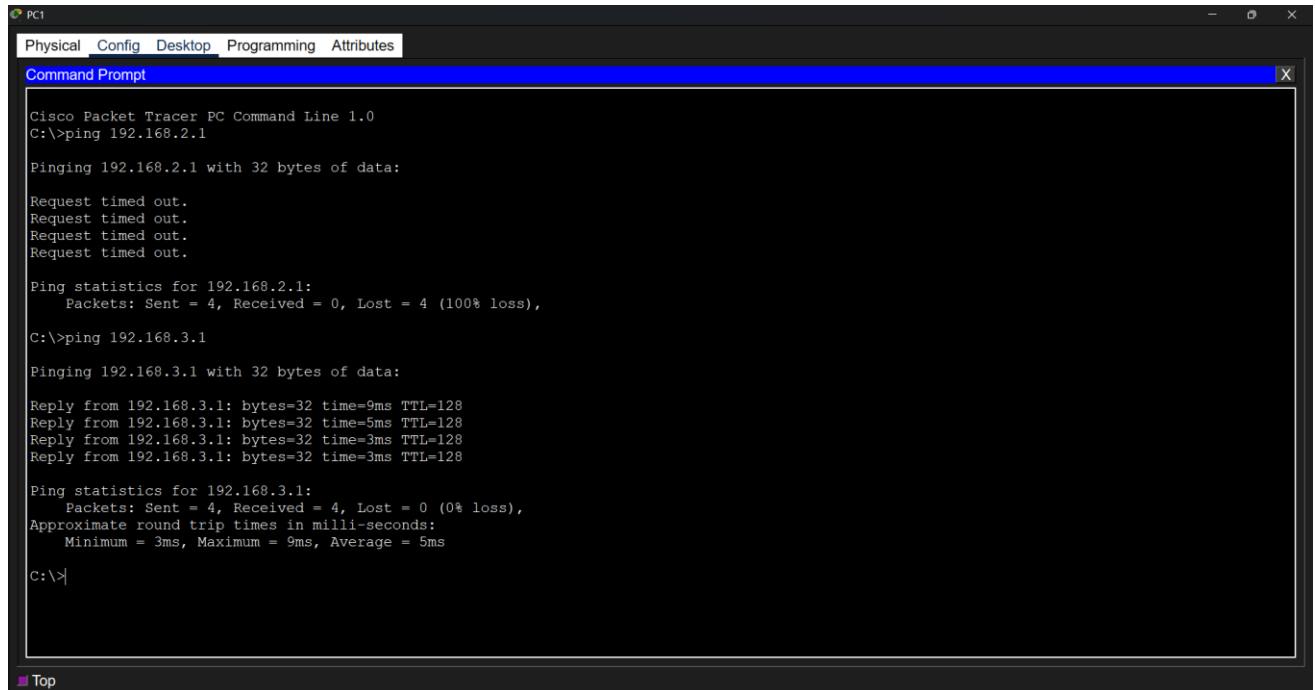
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch(config-if)#swit
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

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3. Verify Connectivity with Ping



The screenshot shows a PC1 window with a Command Prompt open. The prompt displays the results of two ping commands. The first command, 'ping 192.168.2.1', results in four 'Request timed out' messages and a 100% loss. The second command, 'ping 192.168.3.1', results in four successful replies with varying times and a 0% loss.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.3.1

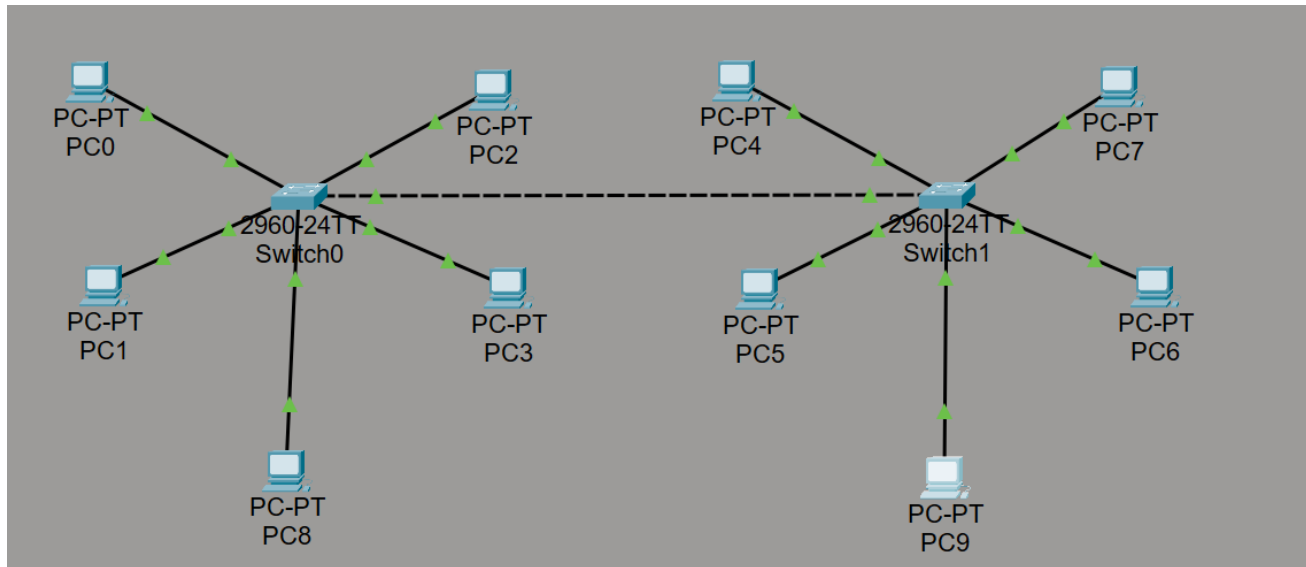
Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time=9ms TTL=128
Reply from 192.168.3.1: bytes=32 time=5ms TTL=128
Reply from 192.168.3.1: bytes=32 time=3ms TTL=128
Reply from 192.168.3.1: bytes=32 time=3ms TTL=128

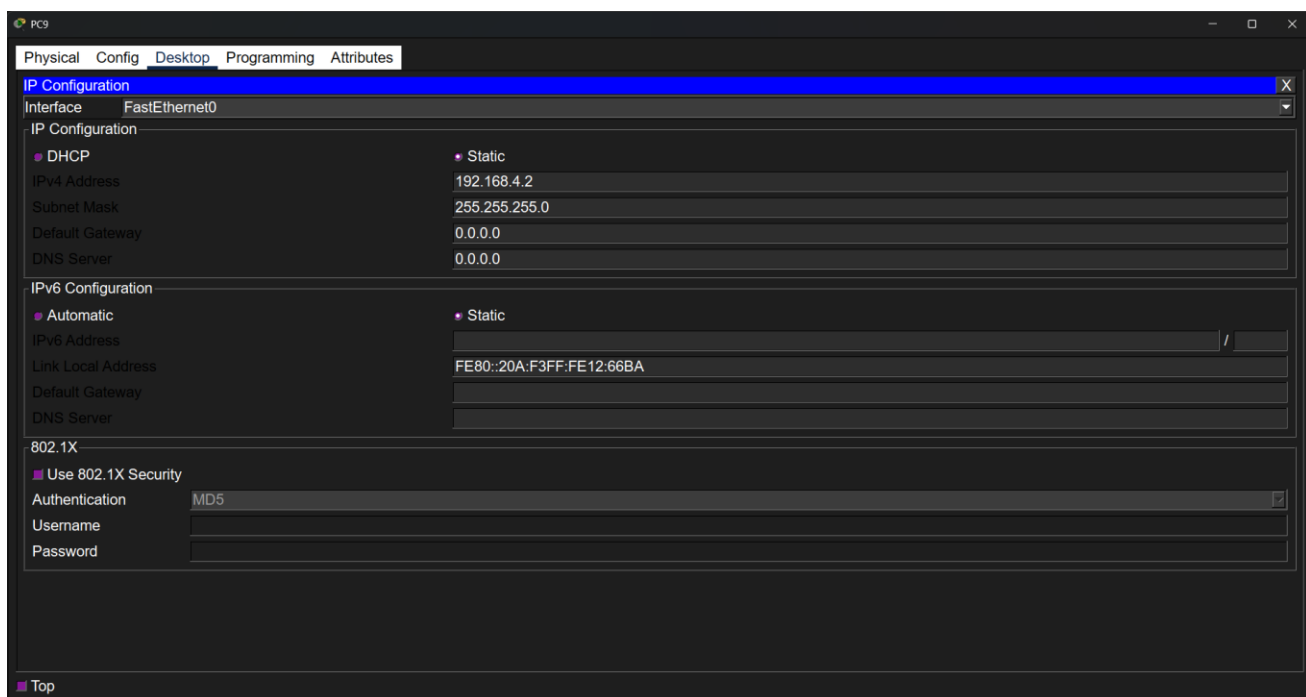
Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 9ms, Average = 5ms

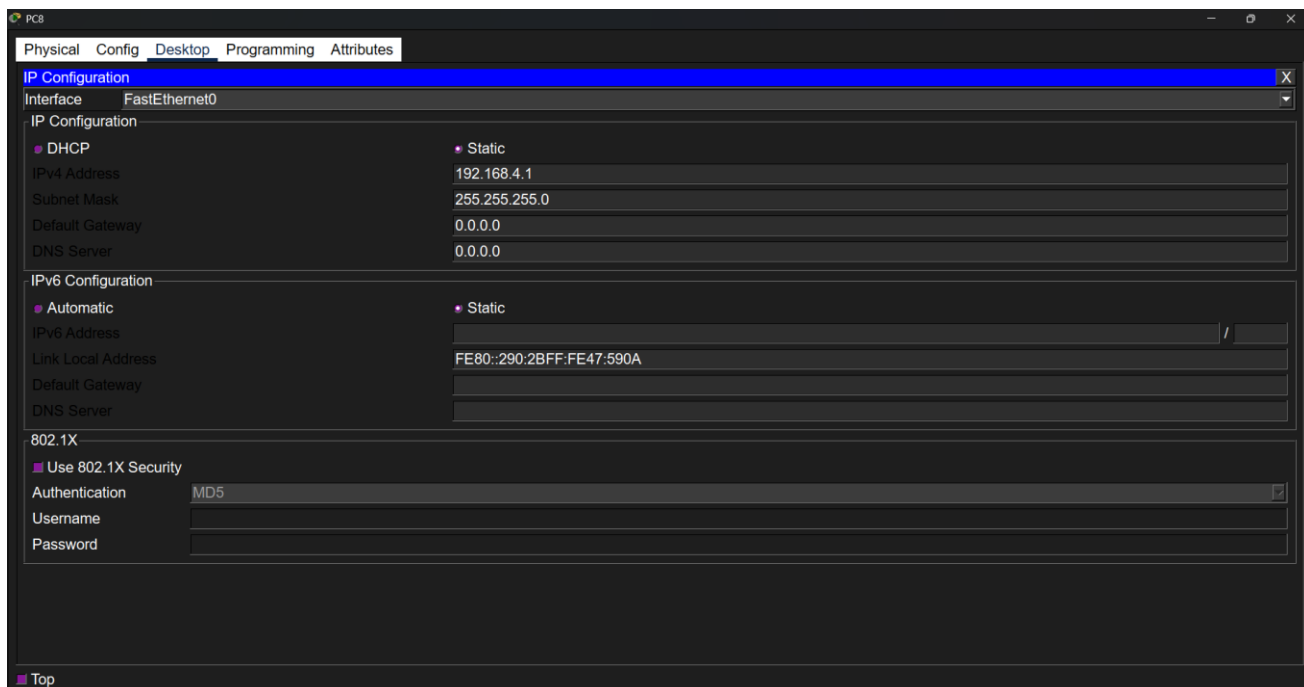
C:\>
```

1.Practice



2. Ip configuration





3. Create Vlan

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5,
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 4
Switch(config-vlan)# name Practical_Task
Switch(config-vlan)#exit
Switch(config)#
```

4. Configure interfaces

```
Switch(config)#interface fastEthernet 0/5
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 4
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)# switchport trunk allowed vlan 2,3,4
Switch(config-if)#end
Switch#wr mem
%SYS-5-CONFIG_I: Configured from console by console
```


5. Verify with ping

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.4.1

Pinging 192.168.4.1 with 32 bytes of data:

Reply from 192.168.4.1: bytes=32 time<1ms TTL=128
Reply from 192.168.4.1: bytes=32 time<1ms TTL=128
Reply from 192.168.4.1: bytes=32 time=4ms TTL=128
Reply from 192.168.4.1: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.4.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>|
```